

eROSITA × DESI × AION-flow — Experiment Pipeline

Living document — keep in sync with the actual pipeline. Last updated: 2026-07-17. Render to PDF with python docs/render_pipeline_pdf.py (writes docs/pipeline.pdf).

1. Goal

Use a frozen astronomical foundation model (AION-1 base) as an information probe: predict eROSITA X-ray properties (broad-band flux, luminosity, hardness) and host stellar mass from non-X-ray DESI modalities (optical spectra, WISE, redshift, Legacy imaging) with a probabilistic normalizing-flow head. Report likelihood gain over a prior, not just point R^2 . This repo extends the PAI26 paper with (a) a cleaned crossmatch and (b) uncertainty-aware, per-target training.

2. Compute

- FASRC** (Harvard SLURM). Account `finkbeiner_lab` (confirmed working 2026-07-21); smokes on `gpu_test`, real runs on `gpu`. `siag_gpu` is *not* visible to us yet — it needs `siag_lab` group membership (Coldfront request pending).
- Env: `module load Miniforge3/26.1.0-fasrc01 + conda env + cu124 torch wheel + pip install -e . polymathic-aion zuko safetensors wandb`. Built at `$AIONFLOW_ROOT/envs/aionflow` (torch 2.6.0+cu124).
- Workflow**: develop locally → `git push` → `git pull` on the cluster; the SSH ControlMaster socket `~/.ssh/cm-fasrc` is used only to submit jobs and move data. All real work runs in `sbatch` jobs on compute nodes (login-node compute gets killed by arbiter2).
- Tracking: Weights & Biases** (`--wandb`, project `erosita-desi-aion-flow`). Logs config, per-step batch NLL, per-epoch train/val NLL, and the final per-combination test metrics table. Auto-falls back to offline if a compute node lacks outbound internet (`wandb sync` later); a tracking failure never kills a job.

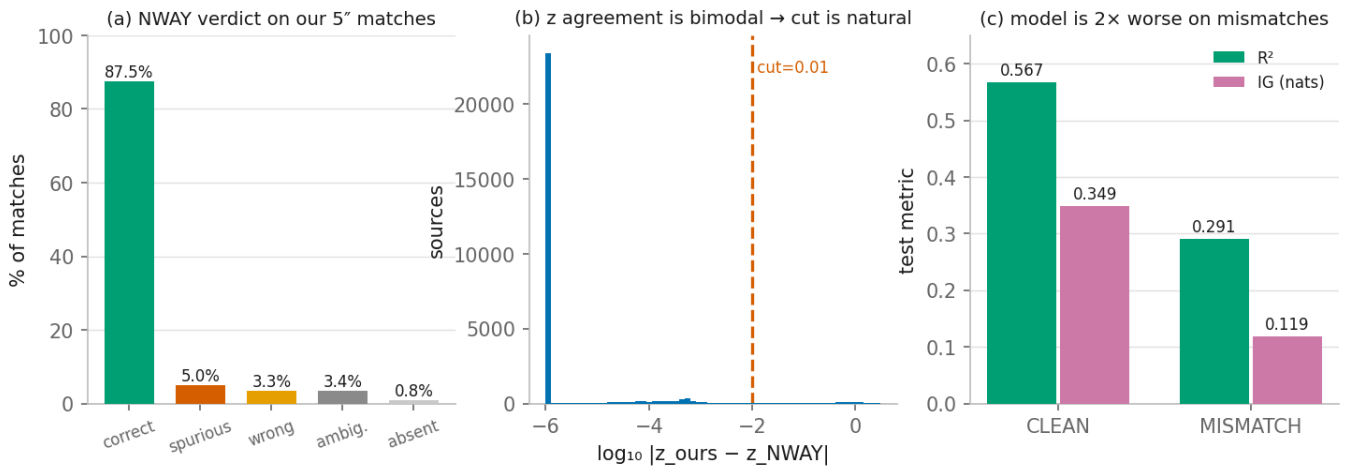
3. Data sources & download

Modality	Source	Notes
X-ray catalog	SRG/eROSITA-DE DR1 / eRASS1 (Merloni+2024)	broad band <code>ML_FLUX_1</code> + sub-bands <code>P1–P4</code> rates/fluxes with asymmetric errors (LOWERR/UPERR). Delivered as the eROSITA×DESI match CSVs.
Optical spectra	DESI DR1 coadd spectra	per-healpix <code>coadd-{survey}-{program}-{healpix}.fits</code> , trimmed to matched targetids, compiled to <code>erosita_spectra_merged_32k.hdf5</code> .
Imaging	Legacy Survey DR10 griz cutouts	160×160 px <code>fits_pool/<targetid>.fits</code> via <code>legacysurvey.org</code> cutout service (~11 GB).
Redshift / galaxy props	DESI DR1 VAC	<code>z</code> , <code>logmstar</code> (photometric), emission lines, <code>spectype</code> (QSO/GALAXY/STAR).
Foundation model	<code>polymathic-ai/aion-base</code> (HuggingFace, ~95 MB)	frozen tokenizer + backbone.

All inputs are local; only `aion-base` is internet-bound. "Adding sources our match missed" would require new DESI DR1 spectrum downloads (deferred).

4. Crossmatch & cleaning

- **Original match (existing):** naive 5" nearest-neighbor (`crossmatch_radec.py`, `astropy`) between DESI target RA/Dec and the eROSITA catalog → 33,938 pairs → 30,441 unique DESI targets → **28,646 with a retrieved spectrum**.
- **Quality problems found:** separations pile up near the 5" cut (median 3.08", 52% >3"); **14% of eROSITA sources match >1 DESI target** (26% of pairs in multi-match groups) → the same X-ray label duplicated onto several spectra.
- **Cleaning (NWAY):** validate against **Salvato et al. 2025 (A&A 704, A344)** — the eRASS1 Bayesian NWAY counterpart catalog — via VizieR J/A+A/704/A344/dr1ls10 (join on IAUName; compare their counterpart's zsp to our matched DESI z). Per-targetid class: **87.5% correct, 4.6% spurious, 3.3% wrong, 3.4% ambiguous, 0.8% absent** → keep = correct filter (`data/match_quality.csv`).
- **Sanity checks (cleaning is sound, 3 independent angles):** (a) $|z_{\text{ours}} - z_{\text{NWAY}}|$ is **bimodal** — 88.7% <1e-4 (same object), 3.0% >0.1 (different), only 0.6% in the 0.01–0.1 valley → the z-cut is natural, not tuned; (b) mismatches show larger match separation (3.19" vs 3.06") but only weakly — separation can't discriminate at eROSITA's ~arcsec position errors, which is *why* prior-based NWAY is needed; (c) the trained model is **2× worse on mismatches** (R^2 0.29 vs 0.57) — independent confirmation the flags localize real failures.
- **Pipeline order (IMPORTANT):** clean → **dedup to the NWAY counterpart per targetid** → **re-derive the train/val/test split on the cleaned sample** (targetid-grouped 80/10/10, seed 42, leakage-safe) → stage. Splits must be redone *after* cleaning.



5. Targets & uncertainties

Four separate 1-D flow runs (`data/targets_extra.csv` carries the derived targets + errors, keyed by `targetid`):

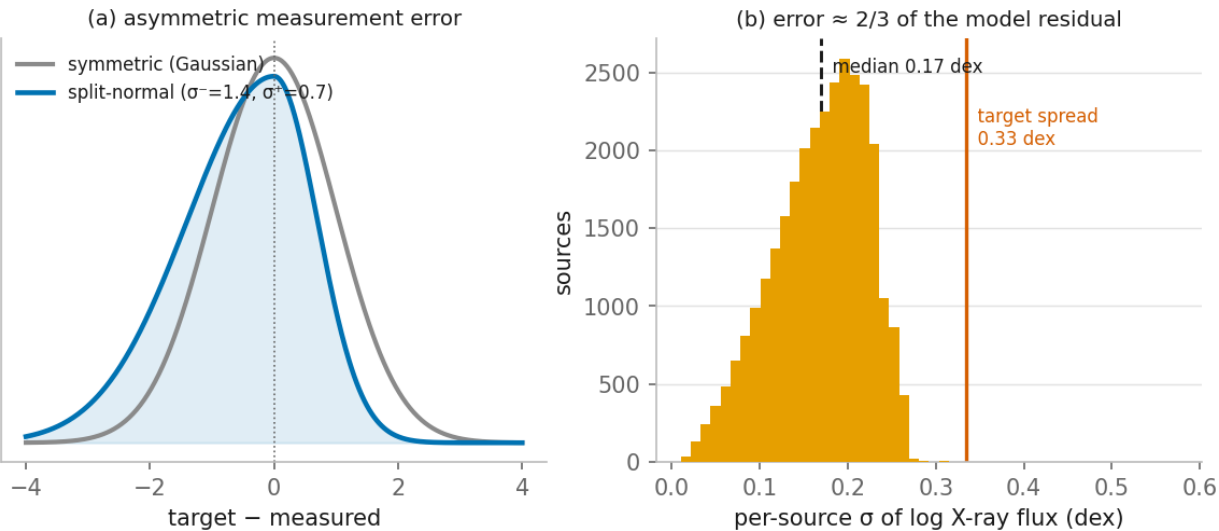
Target	Source	Error model
<code>log_ml_flux_1</code>	<code>log10(ML_FLUX_1)</code>	split-normal (two-piece Gaussian) from LOWERR/UPERR: $\text{sig_hi} = \log_{10}(1 + \text{UPERR}/F)$, $\text{sig_lo} = -\log_{10}(1 - \text{LOWERR}/F)$ (capped). Median ≈ 0.17 dex; 98% asymmetric, low side heavier .
<code>log_lx</code>	<code>ML_FLUX_1 + z</code> (Planck18)	same dex σ as flux.
<code>logmstar</code>	DESI VAC photometric mass	spectype systematic floor : 0.20 dex (GALAXY) / 0.30 dex (QSO). No per-source posterior σ available (PROVABGS covers only 0.4% of this AGN sample). Caveat: 82% of sample is QSO → mass AGN-contaminated; evaluate split by spectype.

hr32 (hardness)	$(P3-P2)/(P3+P2)$ band rates	modeled in arctanh space $u = \text{arctanh}(HR)$, $\text{sig}_u = \text{sig}_{hr} / (1 - HR^2)$. Only ~17% have $S/N > 2$ ($\sigma \approx \text{signal}$) → IG-primary , R^2 secondary. HR43/T dropped.
-----------------	------------------------------	---

Targets & error treatment (four separate V1 runs, all --error-mode convolve)

target	source	error model	mode	trainable
log_ml_flux_1	eROSITA ML_FLUX_1	split-normal (LO/HI), ~0.17 dex	convolve	full sample ✓
log_lx	flux + z (Planck18)	same dex σ as flux	convolve	full (z-dominated)
logmstar	DESI photometric mass	spectype floor 0.2/0.3 dex	convolve	clean for GALAXY; 82% QSO
hr32_u	eROSITA band rates (arctanh)	propagated σ_{HR}	convolve	IG-primary; ~17% $S/N > 2$

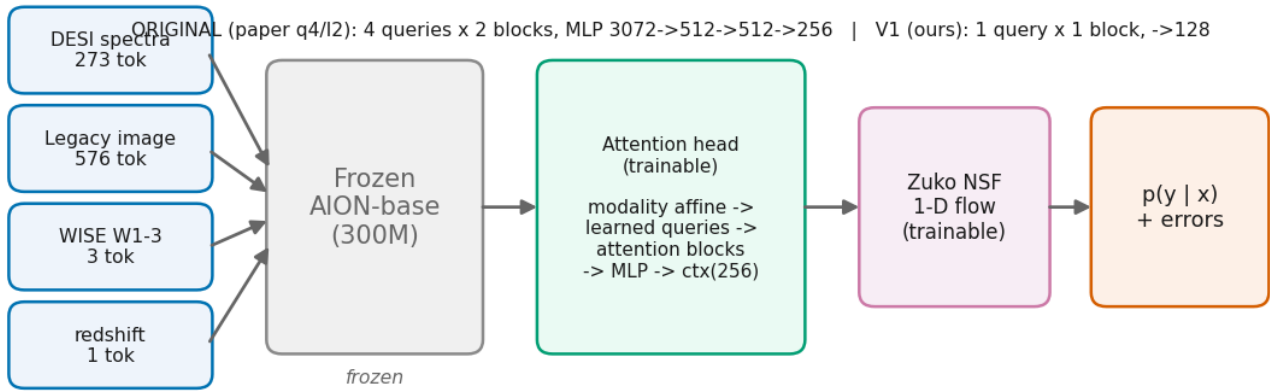
How errors enter the flow (split-normal convolution likelihood): $NLL_i = -\log \int p_{\text{flow}}(t \mid x_i) \cdot K(y_i \mid t; \text{sig}_{\text{lo}_i}, \text{sig}_{\text{hi}_i}) \, dt$, a fixed 1-D quadrature (~21–41 nodes over $\pm 4\sigma$), $\text{error_mode} \in \{\text{none}, \text{inject}, \text{convolve}\}$ (default **convolve** = deconvolves; **inject** broadens; **none** = paper behavior).



Measurement error is large relative to signal: for flux, $\sigma \approx 0.17$ dex is $\sim 2/3$ of the model's residual variance → the model sits near the measurement-noise floor, so error-aware training + a reported error floor matter.

Deferred error refinement: X-ray sub-band non-detections are **upper limits (censored)**, not missing/large-error; proper handling via a censored likelihood (ref 2022A&A...661A...3S). First pass uses split-normal σ on measured values.

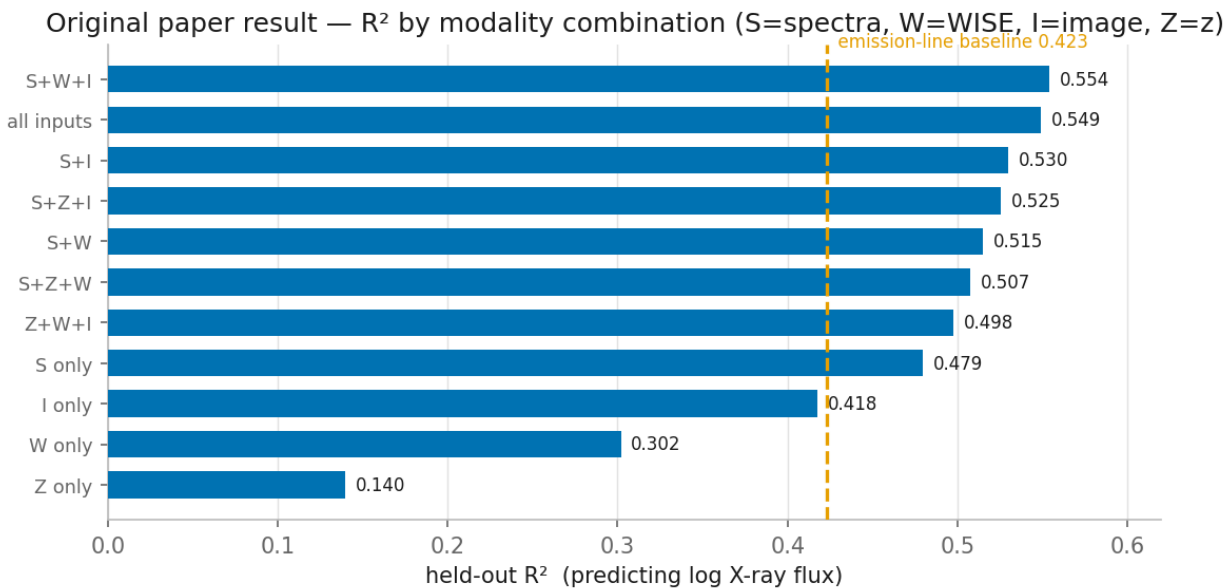
6. Model architecture



Frozen **AION-base** tokenizer + backbone → per-modality token sequences (spectra=273, image=576, WISE=3, z=1; missing modalities omitted). Trainable head + flow:

- **V1 minimal head** (AIONAttentionContext, parameterized): per-modality affine calibration → **1 learned query + 1 attention block (8 heads)** cross/self-attention → 768 → MLP (→128) → 256-d context. (~7.8M params vs the paper's 4-query/3072-d ~16.8M.)
- **Flow**: Zuko conditional **NSF**, 1-D target, 8 transforms, context 256; standardized target; **KDE (Scott)** prior as the IG reference.
- Per-batch a modality combo is sampled; AION frozen; AdamW; checkpoint best all-inputs val NLL.
- **V2 (later)**: replace frozen AION with a self-trained attention encoder over the raw spectrum.

Original paper result (q4/I2, epoch-13) — R^2 by modality combination; full spectra beat the emission-line baseline, all-modality is best:



7. Training & evaluation

- Four separate V1 runs (`--target {log_ml_flux_1, log_lx, logmstar, hr32}`), `--clean` (NWAY keep), `--error-mode convolve`, on gpu; smoke on gpu_test with a `--limit` subset.
- **Metrics**: primary **IG / exp(IG)** vs KDE prior (esp. HR); secondary **R^2 + reduced- χ^2** using per-source σ + the measurement-error floor; **clean-vs-full delta**; per-spectype split for logmstar/HR.
- Every run is mirrored to **W&B** (`--wandb`), so training curves are watchable live and runs are comparable across targets/error-modes via the run tags (<target>, <error_mode>, V1).

8. Key findings so far

- Naive 5" match is **~8% wrong** per NWAY; the paper's q4/l2 model is **~2× worse on the mismatches (R^2 0.29 vs 0.57)**; dropping them lifts test R^2 from the published **0.549** → **0.567** (no retraining).
- X-ray targets are measurement-noise-dominated (flux $\sigma \approx 0.17$ dex; HR $\sigma \approx$ its own spread) → uncertainty handling is essential.

9. Status

- ■ Repo cloned; V1 head coded+tested; FASRC scaffolding + **verify-before-push staging scripts** (scripts/fasrc_preflight.sh, scripts/fasrc_stage_data.sh).
- ■ Sidecars built: data/match_quality.csv (clean filter), data/targets_extra.csv (targets + split-normal/arctanh errors).
- ■ **Error-aware flow** implemented + tested (split_normal_log_kernel, convolve_logprob, ConditionalNSFFlow.log_prob_convolved; 14/14 tests, recovers analytic Gaussian convolution).
- ■ **Clean** → **dedup** → **re-split** implemented + tested (build_clean_manifest; 25,200 clean rows, exact 80/10/10, leakage-safe, deterministic).
- ■ **Full wiring done + validated locally**: prepare-data --clean cleans/dedups/re-splits + writes extra target & error columns; AIONHDF5Dataset emits per-source σ (10-tuple); main.py --target {..., logmstar, hr32_u} / --error-mode {none, inject, convolve} / --clean; batch_nll uses log_prob_convolved. 14/14 tests; a real prepare-data --clean --limit 12 produced correct columns.
- ■ **FASRC ready**: 17 GB data staged + fits_pool unzipped (10,355 cutouts); conda env built and verified (torch 2.6.0+cu124, aion, zuko all import); repo is a real git clone of multitarget-clean-errors; account finkbeiner_lab confirmed via sbatch --test-only.
- ■ **W&B tracking** wired into train (opt-in --wandb, offline-safe, no-op if absent). First live run confirmed end-to-end on gpu_test.
- ■ **Staged-data validation** (scripts/validate_staged.py + 13 tests + sbatch/validate_staged.sbatch), wired as a hard gate in train.sbatch: schema, row-count alignment, dedup/leakage/split-fractions, NWAY clean filter, σ positivity & coverage, value ranges, fits coverage, and the dataloader 10-tuple contract.
- ■■ **Caught by that validation**: the fits_pool unzip had been interrupted (10,355 of 27,373 cutouts on disk). Because staging drops any object lacking a cutout, the first full run staged **9,592 rows instead of ~22k** with no error. Fixed (count-based unzip idempotency + a coverage check that fails above 20% loss); re-staging.
- ■ **Next (reproduction-first ladder — every change we made is flag-gated with paper defaults, so the baseline runs from this branch with no rollback)**: 1. finish re-staging (staged/ cleaned+re-split, staged_paper{, _smoke}/ = original manifest, no clean, no error columns) + validate both; 2. **repro smoke** on gpu_test (paper q4/l2 head, --error-mode none, staged_paper_smoke); 3. **paper-table reproduction** on gpu (50 epochs, batch 448, paper config exactly) → compare to the published table (all-inputs R^2 0.549 expected; our re-staged sample may differ slightly from the paper's exact rows — the provenance report quantifies this); 4. then the new-science runs (cleaned + convolve, V1 head) — the clean-vs-paper delta is the first result.
- ■■ siag_lab membership pending (Coldfront) → siag_gpu invisible for now; using gpu_test (smokes) / gpu (real runs) under finkbeiner_lab.
- Known refinement: hr32_u piles at the arctanh clip for low-S/N HR (huge σ) — fine for flux/lx/logmstar; the HR run wants an S/N gate/clip.

10. Deferred / out of scope

SFR (needs DESI FastSpecFit VAC), Γ (=HR relabel), X-ray upper-limit censoring, full Salvato re-crossmatch of "the rest", V2 self-trained encoder, saliency/attribution.